

Tensor Layout Mastery Guide

A highly structured reference manual covering multidimensional data representation, slicing, indexing, and broadcasting operations in deep learning.

1. Image Representation

A grayscale image has a resolution of 28×28 pixels. How can it be represented as a tensor?

- A. Scalar
- B. 1D tensor of shape (28,)
- C. 2D tensor of shape (28, 28)
- D. 3D tensor of shape (1, 28, 28)

Correct Answer: C. 2D tensor of shape (28, 28)

2. RGB Image

An RGB image has a resolution of 224×224 pixels. Which tensor shape is most appropriate for a single image in a deep-learning framework using channels-last representation?

- A. (3, 224, 224)
- B. (224, 224, 3)
- C. (224, 3, 224)
- D. (224, 224)

Correct Answer: B. (224, 224, 3)

3. Batch of Images

Suppose we have 32 RGB images, each of size 224×224 . Using channels-last representation, what is the tensor shape?

- A. (224, 224, 3)
- B. (32, 224, 224, 3)
- C. (3, 224, 224, 32)
- D. (32, 3)

Correct Answer: B. (32, 224, 224, 3)

4. What Does 32 Mean?

Consider the tensor: $X.shape = (32, 224, 224, 3)$. What does 32 most likely represent?

- A. Number of pixels
- B. Number of channels
- C. Batch size
- D. Image width

Correct Answer: C. Batch size

5. What Does 3 Mean?

For: $X.shape = (32, 224, 224, 3)$. What does the final dimension represent?

- A. Number of images
- B. Image height
- C. Image width
- D. Number of color channels

Correct Answer: D. Number of color channels

6. Tensor Rank

Consider: $X.shape = (32, 224, 224, 3)$. What is the rank/number of axes of this tensor?

- A. 3
- B. 4
- C. 32
- D. 224

Correct Answer: B. 4

7. Batch Dimension

Suppose a neural network expects input of shape: $(batch_size, height, width, channels)$. Which of the following represents 16 RGB images of size 128×128 ?

- A. $(16, 128, 128, 3)$
- B. $(128, 128, 16, 3)$
- C. $(3, 128, 128, 16)$
- D. $(16, 3, 128)$

Correct Answer: A. $(16, 128, 128, 3)$

8. Sensor Data

An industrial system records data from 10 sensors, once every second, for 60 seconds. If we store the data as: $X.shape = (60, 10)$. What does each row represent?

- A. One sensor
- B. One time instant
- C. One complete experiment
- D. One feature

Correct Answer: B. One time instant

9. Adding Multiple Machines

Suppose the same 60-second experiment is performed on 100 machines. If the data is represented as: $X.shape = (100, 60, 10)$. What do the three axes represent?

- A. Machines $\times$ Time $\times$ Sensors
- B. Time $\times$ Machines $\times$ Sensors
- C. Sensors $\times$ Machines $\times$ Time
- D. Machines $\times$ Sensors $\times$ Time

Correct Answer: A. Machines $\times$ Time $\times$ Sensors

10. Text as a Tensor

A sentence is converted into token IDs: [15, 42, 87, 102, 9]. What is the simplest tensor representation?

- A. Scalar
- B. 1D tensor
- C. 2D tensor
- D. 3D tensor

Correct Answer: B. 1D tensor

11. Understanding the Embedding Dimension

In: $X.shape = (32, 100, 768)$. What does 768 represent?

- A. Vocabulary size
- B. Number of sentences
- C. Number of tokens
- D. Embedding dimension

Correct Answer: D. Embedding dimension

12. Tensor Indexing

Consider: $X.shape = (32, 100, 768)$. What does $X[0]$ represent?

- A. The first token across all sentences
- B. The first embedding dimension
- C. The first sentence/sequence
- D. The first batch dimension only

Correct Answer: C. The first sentence/sequence

13. Tensor Slicing

For: $X.shape = (32, 100, 768)$. What would $X[:, 0, :]$ select?

- A. The first sentence
- B. The first token from every sentence
- C. The first embedding dimension from every token
- D. The entire tensor

Correct Answer: B. The first token from every sentence

14. Shape Compatibility

Suppose: $A.shape = (32, 100, 768)$ and $B.shape = (32, 100, 768)$. Which operation is naturally possible element-by-element?

- A. Addition
- B. Only matrix multiplication
- C. Only concatenation
- D. No operation is possible

Correct Answer: A. Addition

15. Broadcasting

Suppose: $X.shape = (32, 100, 768)$ and $b.shape = (768,)$. What can happen when b is added to X ?

- A. Addition is impossible because the shapes are different
- B. b can be broadcast across the first two dimensions
- C. X is automatically reduced to $(768,)$
- D. The batch dimension disappears

Correct Answer: B. b can be broadcast across the first two dimensions

16. Concatenation

You have two feature tensors: $A.shape = (32, 100, 128)$ and $B.shape = (32, 100, 256)$. You concatenate them along the feature dimension. What is the resulting shape?

- A. $(32, 100, 384)$
- B. $(32, 200, 384)$
- C. $(64, 100, 384)$
- D. $(32, 100, 256)$

Correct Answer: A. $(32, 100, 384)$

17. Reshaping

A tensor has shape: (4, 5, 6). How many total elements does it contain?

- A. 15
- B. 60
- C. 120
- D. 24

Correct Answer: C. 120

18. Valid Reshape

A tensor has shape: (4, 5, 6). Which reshape is valid without changing the number of elements?

- A. (4, 10)
- B. (2, 30)
- C. (3, 20, 2)
- D. Both B and C

Correct Answer: D. Both B and C

19. Reshape vs Transpose

An engineer changes a tensor from: (32, 224, 224, 3) to: (32, 3, 224, 224). Which operation is conceptually appropriate?

- A. Reshape
- B. Transpose/permute axes
- C. Flatten
- D. Aggregation

Correct Answer: B. Transpose/permute axes

20. Why Does This Matter?

A deep-learning model expects: (batch, channels, height, width) but the image data is stored as: (batch, height, width, channels). What is the main issue?

- A. The numerical values are necessarily wrong
- B. The model may interpret the dimensions incorrectly
- C. The image has too many pixels
- D. The batch size becomes zero

Correct Answer: B. The model may interpret the dimensions incorrectly

21. Flattening

An RGB image has shape: (224, 224, 3). If flattened into a 1D tensor, how many elements will it contain?

- A. 224
- B. 448
- C. 150,528
- D. 50,176

Correct Answer: C. 150,528

22. Tensor vs Matrix

Which statement is most accurate?

- A. A tensor is always a 3D object
- B. A matrix is a special case of a tensor
- C. A tensor cannot contain numerical values
- D. A matrix and tensor are completely unrelated concepts

Correct Answer: B. A matrix is a special case of a tensor

23. Neural Network Activations

A neural network processes a batch of 64 samples. Each sample produces a feature vector of length 128. What is the likely shape of the activation tensor?

- A. (128,)
- B. (64,)
- C. (64, 128)
- D. (128, 128)

Correct Answer: C. (64, 128)

24. Convolutional Layer

A CNN receives: $X.shape = (16, 3, 224, 224)$ and produces: $Y.shape = (16, 64, 112, 112)$. What has most likely changed?

- A. Batch size only
- B. Number of channels and spatial resolution
- C. Only the number of samples
- D. Vocabulary size

Correct Answer: B. Number of channels and spatial resolution

25. Tensor as Data Organization

Why are tensors particularly useful in deep learning?

- A. They provide a structured way to represent multidimensional data and enable efficient numerical operations
- B. They eliminate the need for mathematics
- C. They automatically understand the meaning of every dimension
- D. They are only required for images

Correct Answer: A. They provide a structured way to represent multidimensional data and enable efficient numerical operations

26. Rank vs Size

A tensor has shape: (1000, 224, 224, 3). An engineer says: 'This is a rank-1000 tensor because the first dimension is 1000.' Is the statement correct?

- A. Yes
- B. No, its rank is 4; 1000 is the size of its first axis
- C. Yes, because rank equals the largest dimension
- D. No, because tensors cannot have rank 4

Correct Answer: B. No, its rank is 4; 1000 is the size of its first axis

27. Challenge Question

Consider: $X.shape = (8, 30, 12, 64)$. Which interpretation is most reasonable for a sequence-based engineering application?

- A. 8 samples, 30 time steps, 12 sensors, 64 features per sensor
- B. 8 sensors, 30 samples, 12 features, 64 time steps
- C. 8 features, 30 sensors, 12 samples, 64 batches
- D. The shape alone tells us the exact semantic meaning of every dimension

Correct Answer: A. 8 samples, 30 time steps, 12 sensors, 64 features per sensor